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Audiovisual distraction for pain relief in paediatric inpatients: A crossover study

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Abstract

Background: Pain is a stressful experience that can have a negative impact on child development. The aim of this crossover study was to examine the efficacy of audiovisual distraction for acute pain relief in paediatric inpatients.
Method: The sample comprised 40 inpatients (6–11 years) who underwent painful puncture procedures. The participants were randomized into two groups, and all children received the intervention and served as their own controls. Stress and pain-catastrophizing assessments were initially performed using the Child Stress Scale and Pain Catastrophizing Scale for Children, with the aim of controlling these variables. The pain assessment was performed using a Visual Analog Scale and the Faces Pain Scale-Revised after the painful procedures. Group 1 received audiovisual distraction before and during the puncture procedure, which was performed again without intervention on another day. The procedure was reversed in Group 2. Audiovisual distraction used animated short films. A 2 × 2 × 2 analysis of variance for 2 × 2 crossover study was performed, with a 5% level of statistical significance.
Results: The two groups had similar baseline measures of stress and pain catastrophizing. A significant difference was found between periods with and without distraction in both groups, in which scores on both pain scales were lower during distraction compared with no intervention. The sequence of exposure to the distraction intervention in both groups and first versus second painful procedure during which the distraction was performed also significantly influenced the efficacy of the distraction intervention.
Conclusion: Audiovisual distraction effectively reduced the intensity of pain perception in paediatric inpatients.
Significance: The crossover study design provides a better understanding of the power effects of distraction for acute pain management. Audiovisual distraction was a powerful and effective non-pharmacological intervention for pain relief in paediatric inpatients. The effects were detected in subsequent acute painful procedures.

Pain constitutes a global health problem, the relief and treatment of which are human rights that are recognized by the World Health Organization and International Association for the Study of Pain (2008). Despite the exponential increase in scientific evidence on paediatric pain in recent decades, many barriers have been faced in translating scientific knowledge into clinical practice. Consequently, children continue to experience pain unnecessarily during hospitalization (Karling et al., 2002; Harrison et al., 2006; Taylor et al., 2008; Stevens et al., 2011; Linhares et al., 2012; Birnie et al., 2014a).

Recent scientific evidence shows that the pain that is associated with acute painful procedures (e.g. venipuncture and immunization) is a major source of paediatric pain and may have long-term consequences on behaviour and the perception of pain in children (Howard, 2003; Kennedy et al., 2008; Noel et al., 2012). Non-pharmacological management, including strategies that are based on distraction, have been shown to have a positive impact on pain relief during acute painful procedures (Kleiber and Harper, 1999; Uman et al., 2008, 2013; Birnie et al., 2014b). Distraction is a cognitive resource that seeks to divert the child's attention away from aversive aspects of painful procedures and towards more pleasant stimuli to reduce the impact of these experiences with regard to fear and affective responses to painful experiences (Chen et al., 2000; Linhares and Doca, 2010; Miller et al., 2011). Distraction interventions involve imagination, engagement and motivation, and the child can learn to build his own distraction using different types of distractors (e.g. books, video games and movies).

Previous reviews have described some methodological shortcomings of non-pharmacological interventions for pain relief, such as few randomized controlled trials, small sample sizes and the lack of accurate descriptions of the procedure for pain relief (Stinson et al., 2008; Uman et al., 2008, 2013; Hermann, 2012). A recent review highlighted the need to improve the methodological quality of studies in the area to develop guidelines for clinical practice, highlighting the importance of evaluating the impact of types of distraction and sample characteristics (Birnie et al., 2014b).

Further randomized controlled trials are needed, which may provide evidence of the efficacy of distraction techniques for non-pharmacological pain management. To our knowledge, no studies have used a crossover design to evaluate only distraction interventions for pain relief with paediatric inpatients who serve as their own controls. Such study designs will advance knowledge of the power of the effect of distraction interventions, and two conditions can be tested using the same participant, thus reducing individual confounding variables. Considering this gap in the literature, this crossover study evaluated the efficacy of audiovisual distraction for the relief of acute pain in hospitalized children who underwent painful clinical procedures while controlling for stress and pain-catastrophizing variables.

1. Methods

1.1 Sample

The study sample consisted of 40 children of school age (6–11 years) who were admitted to a paediatric ward and subjected to venipuncture or arterial puncture procedures based on clinical demand. This procedure was indicated by paediatricians who provided care for the children. The procedures were performed by trained nurses. The sample calculation indicated that 15 participants in each group would be satisfactory to determine the effect of the intervention, with statistical power of 0.80 at a level of significance of 5% ($p \leq 0.05$) and two points reduction on the Faces Pain Scale-Revised, which goes up to 10 points.

The participants were randomized into two groups. All of the children received the distraction intervention at two different times and served as their own controls. The randomization procedure was performed according to the website www.randomizer.org (accessed April 1, 2015). Group 1 (G1, $n = 22$) consisted of children who received the distraction intervention during the first painful procedure (A) and then underwent the usual hospital routine during the second painful procedure on another day that consisted of no non-pharmacological intervention for this specific procedure (B). Group 2 (G2, $n = 18$) consisted of children who underwent the usual hospital routine during the first painful procedure (B) and then received the distraction intervention during the second painful procedure on another day (A). The painful procedures occurred at time intervals of at least 1 day and a maximum of 7 days. This time interval between puncture procedures was established to guarantee subject participation in the study, reduce the dropout rate and control possible changes in the clinical status of the paediatric inpatients.

The inclusion criteria for the participants were the following: children who were aged 6–12 years, had sufficient language skills to communicate, were in a hospital setting with different diagnoses, were prescribed venipuncture or arterial puncture based on clinical demand, were hospitalized for a minimum of 1 day and had a time interval of not more than 7 days between the puncture procedures. The exclusion criteria were the following: children with neurological impairments or communication problems and children under sedation.

During the data collection period (March–November 2013), 177 children (6–12 years of age) were admitted to the paediatric ward. Of these children, 135 did not meet the inclusion criteria for the study or met any of the exclusion criteria. The predominant reason for exclusion from the study was our difficulty in planning two puncture procedures at a maximum time interval of 7 days. Other children were excluded because of neurological impairments or communication problems, and some children were under sedation. Of the remaining 42 children, one family refused to participate in the study because of their religious beliefs, and one child did not complete the study because he was discharged early.

1.2 Ethical aspects

The study was approved by the Research Ethics Committee of the Clinics Hospital, Ribeirão Preto Medical School, University of São Paulo, Brazil (Hospital das Clínicas da Faculdade de Medicina de Ribeirão Preto da Universidade de São Paulo; HCFMRP-USP).

1.3 Instruments

1.3.1 Visual analogue scale

A visual analogue scale (VAS; Scott et al., 1977) was used for the assessment of pain intensity during the procedure. The horizontal VAS consisted of a 10 cm horizontal line that ranged from 0 (no pain) to 10 (most pain). The patients were asked to rate their pain on the horizontal VAS after the painful procedure. The instrument has strong psychometric qualities (Dworkin et al., 2005; von Baeyer, 2006; McGrath et al., 2008).

1.3.2 Faces Pain Scale-Revised

The Brazilian Portuguese version of the Faces Pain Scale-Revised (FPS-R; Hicks et al., 2001) is validated for the Brazilian population (Poveda et al., 2001; Charry et al., 2014). The revised scale consists of six faces, with a metric correlation of 0–10. The instrument measures pain intensity and presents strong psychometric qualities (Dworkin et al., 2005; von Baeyer, 2006; McGrath et al., 2008).

1.3.3 Pain-catastrophizing scale for children

The Pain Catastrophizing Scale for Children (PCS-C) consists of 13 items on a 5-point scale that are based on the frequency that children experience thoughts and feelings when they are in pain ("not at all," "mildly," "moderately," "severely" and "extremely"). Scores range from 0 to 52. The PCS-C has three subscales that are related to magnification, rumination and helplessness dimensions. The version for children was developed and validated by Crombez et al. (2003) and has construct validity, internal validity and predictive validity in school children aged 8 and older. We obtained authorization from the authors to translate the scale into Brazilian Portuguese for use in this study. We performed a translation procedure that included the following steps. First, the instrument was translated into Brazilian Portuguese by the first author. The translation was reviewed by the last author. Both investigators are native Brazilian Portuguese speakers and fluent in English. Second, we backtranslated the instrument with the aid of a native English speaker who is fluent in Brazilian Portuguese. The two versions of the instrument were then compared to confirm the meaning of all of the items. After obtaining the final Brazilian Portuguese version, the instrument was applied in five children who were similarly aged as the children in this study to determine whether the instrument was understandable by the children.

1.3.4 Child Stress Scale

The Child Stress Scale (CSS; Lipp and Lucarelli, 1998) is composed of 35 items that are related to stress reactions that may present in four dimensions: physical, psychological, psychological with depression component and psychophysiological. Scores range from 0 to 140. Following are examples of the items: "I worry that bad things can happen"; "I feel scared by the time I go to sleep" and "I get nervous with everything." The following responses were possible: "I never feel"; "I rarely feel"; I sometimes feel"; "I frequently feel" and "I always feel."

2. Materials

To conduct the distraction intervention, we used a portable Philips PD7030 DVD player with a 7-inch LCD screen to play the Pixar Short Films Collection, Volume 1, which contains children's stories with characters produced by Walt Disney Pictures.

2.1 Procedure

2.1.1 Data collection

The CSS and PCS-C were first applied. Information about the painful procedure was clearly given to the children so that they would understand what would happen during the procedure. At the time of the study, no non-pharmacological (e.g. guided imagery and psychological preparation) or pharmacological protocols (e.g. topical anaesthetic cream for pain relief) were adopted by the hospital in the routine puncture procedures.

The audiovisual distraction intervention consisted of directing the child's attention to the short films before and during the painful procedure. The child began to watch the film 5 min before the puncture procedure with the support of the researcher. While the children watched the short films, the researcher guided the children's attention to the content of the films, making comments about the scenes and characters. The film was presented for an average of 10 min to cover the entire length of the puncture procedure. The child continued to watch the movie until it ended through the recovery phase of the procedure. The parents of the children were present during the intervention, but they were instructed to not interfere with the procedure.

Group 1 (G1) received the distraction intervention (A), which was conducted by the principal investigator (experienced and trained psychology specialist in Pediatric Psychology) before and during the puncture procedure. On another day, a second puncture procedure was performed without the distraction intervention (B) with the same children who were previously assessed. In Group 2 (G2), the sequence was reversed. The participants were first subjected to the puncture procedure without the distraction intervention (B), and the painful procedure was performed again on another day with the distraction intervention (A). Immediately after the painful procedures, the researcher applied the pain intensity instruments (VAS and FPS-R).

Notes were taken for each painful procedure, including the type of puncture (venous or arterial) and number of attempts to achieve a successful puncture. The following information was also collected from the medical records: paediatric specialty that was responsible for care of the child, diagnosis, treatment time, current reasons for hospitalization, length of stay until the day of the first puncture, time interval between puncture procedures and total length of hospital stay.

2.1.2 Data analysis

Descriptive statistical analyses were first performed. Between-group comparisons were performed using Student's *t*-test for continuous variables and the χ^2 test for categorical variables. Associations between pain-catastrophizing and pain intensity outcomes were examined using the Pearson correlation test. A $2 \times 2 \times 2$ analysis of variance (ANOVA) for 2×2 crossover study was performed. An Omnibus Measure of Separability was applied to assess carryover effects. Different designs vary in their ability to separate treatment and carryover effects. A measure of separability may be calculated for each design using the observed frequencies of carryover and comparing them to expected frequencies in an independent model. The Omnibus measure of separability of the treatment and carryover effects indicates design separability (Ratkowsky et al., 1993). The Omnibus Measure of Separability is defined as $S = 100$ $(1 - V)$ %, where V is Cramer's measure of association (Cramer's V). The value of S was 30%, meaning that 30% of the treatment and carryover effects could not be separated.

The Omnibus measure was executed using Stata Statistical Software (Stata Corp LP, 13.0, College Station, TX, USA), and the other statistical tests were performed using the Statistical Package for the Social Sciences (SPSS, version 22.0, Chicago, IL, USA). The level of significance for all tests was 5% ($p \leq 0.05$).

3. Results

3.1 Sample characteristics

Table 1 shows no significant differences in the sociodemographic characteristics of the children between G1 and G2. A similar distribution of boys and girls was found in both groups. The children in the two groups were an average of 8 years old, with average schooling of 2–3 years.

Table 1 also reveals no significant differences in the clinical characteristics that were collected from medical records between G1 and G2. The sample consisted of patients from different paediatric specialties, but the majority of children were inpatients in onco-haematology. The current reasons for hospitalization were similar between groups.

The duration of disease status that required treatment was an average of 40 months in both groups, suggesting the presence of chronic disease. The length of hospital stay until the first assessment was an average of 4 days, and the total length of hospital stay was 10 and 11 days in G1 and G2, respectively, with no statistically significant difference between groups. The interval between pain assessments was an average of 2 days in both groups, with no statistically significant difference between groups.

Table 1 Sample characteristics.

Characteristics of the children	Group 1 (AB) (*n* = 22)	Group 2 (BA) (*n* = 18)	*p*
Gender (*n* [%])			
Boys	7 (32)	9 (50)	0.24[a]
Girls	15 (68)	9 (50)	
Age (years) (mean [SD])	8.32 (±2.05)	8.72 (±1.80)	0.52[b]
Education (years) (mean [SD])	2.82 (±1.81)	3.61 (±2.00)	0.20[b]
Paediatric specialty (*n* [%])			
Onco-haematology	12 (54)	5 (29)	NA
Gastroenterology	1 (5)	0	
Cardiology	1 (5)	0	
Endocrinology	1 (5)	5 (29)	
Pulmonology	3 (13)	3 (17)	
Rheumatology	1 (5)	2 (11)	
Nephrology	3 (13)	1 (7)	
Immunology	0	1 (7)	
Current hospitalization (*n* [%])			
Clinical treatment	17 (77)	9 (50)	0.08[a]
Surgical treatment	3 (14)	0	
Diagnostic investigation	2 (9)	9 (50)	
Duration of disease (months) (mean [SD])	40.27 (±43.21)	39.72 (±40.80)	0.97[b]
Hospital stay until first assessment (days) (mean [SD])	4.36 (±4.07)	4.61 (±4.48)	0.86[b]
Total time of hospital stay (days) (mean [SD])	11.77 (±9.71)	10.56 (±7.70)	0.67[b]
Time interval between assessments (days) (mean [SD])	2.32 (±1.64)	2.50 (±1.72)	0.74[b]

SD, standard deviation; NA, not applicable.
[a]χ^2 test.
[b]Student's *t*-test.

3.2 Characteristics of the painful procedure

Table 2 shows that both groups were similar with regard to the characteristics of the painful procedures, with no statistically significant difference between groups. The predominant type of procedure in both groups was venipuncture, followed by arterial puncture, for both the first and second pain assessments. With regard to the number of attempts to achieve a successful puncture, both groups had a similar average of one attempt for both procedures. Most of the children did not receive pain medications on the days of assessment, with no statistically significant difference between groups.

Table 2 also shows no significant differences in the children's pain scores on the FPS-R and VAS between the two types of puncture procedure (venipuncture and arterial) at the first pain assessment (FPS-R, $p = 0.69$; VAS, $p = 0.79$) and second pain assessment (FPS-R, $p = 0.48$; VAS, $p = 0.73$) during the period of the distraction intervention.

3.3 Indicators of stress and pain catastrophizing in children

Table 3 shows no statistically significant differences in scores on the CSS and PCS-C between groups, indicating that the groups were comparable in these variables. However, a statistically significant difference in magnification subscale scores on the PCS-C was found between groups, with higher scores in G2 compared with G1.

Table 3 also reveals high-magnification subscale scores on the PCS-C in G2. Thus, we examined the correlation between this variable and pain scores using both scales. The results showed no significant associations between these variables in the distraction period (FPSR: $r = 0.30$, $p = 0.23$; VAS: $r = 0.43$, $p = 0.78$) and in the no intervention period (FPSR: $r = 0.22$, $p = 0.38$; VAS: $r = 0.18$, $p = 0.48$).

3.4 Indicators of pain intensity

Table 4 presents that the mean scores on the pain scales with the distraction intervention in G1 and G2 were lower than without the intervention. The pain scores in G1 without the distraction intervention were significantly lower compared with G2. Notably, the children in G1 received the audiovisual distraction intervention during the first puncture procedure, and the children in G2 received the distraction intervention during the second puncture procedure.

3.5 Efficacy of distraction intervention for pain relief

3.5.1 Pain outcome in FPS-R measure

A statistically significant difference was found between the periods with and without the distraction intervention. A significant effect of the intervention on pain relief was observed, with lower scores on the FPS-R with the distraction intervention compared with no intervention ($F_{1,38} = 34.27$, $p < 0.001$). The results also showed that the sequence of distraction exposure (AB vs. BA) in both groups ($F_{1,38} = 7.79$, $p = 0.01$) and the first versus second painful procedure ($F_{1,38} = 5.71$, $p = 0.02$) during which the distraction intervention was performed also significantly influenced the efficacy of audiovisual distraction. There were no

Table 2 Characteristics of the painful procedures.

Characteristics of painful procedure	Group 1 (AB) (*n* = 22)	Group 2 (BA) (*n* = 18)	*p*
Type of painful procedure (first assessment) (*n* [%])			
Venipuncture	19 (86)	15 (83)	0.79[a]
Arterial puncture	3 (14)	3 (17)	
Type of painful procedure (second assessment) (*n* [%])			
Venipuncture	20 (91)	16 (89)	0.83[a]
Arterial puncture	2 (9)	2 (11)	
Number of attempts (first assessment) (mean [SD])	1.27 (±0.70)	1.56 (±0.92)	0.28[b]
Number of attempts (second assessment) (mean [SD])	1.18 (±0.66)	1.00 (±0.00)	0.25[b]
Medication used for pain during the day (first assessment) (*n* [%])			
No	14 (64)	12 (67)	0.84[a]
Yes	8 (36)	6 (33)	
Medication used for pain during the day (second assessment) (*n* [%])			
No	13 (59)	14 (78)	0.21[a]
Yes	9 (41)	4 (22)	

SD, standard deviation.
[a]χ^2 test.
[b]Student's *t*-test.

Table 3 Indicators of stress and pain catastrophizing.

Indicators of stress and pain catastrophizing	Group 1 (AB) (*n* = 22)	Group 2 (BA) (*n* = 18)	*p*
Presence of clinical stress in children (*n* [%])			
No	21 (95)	15 (83)	0.20[a]
Yes	1 (5)	3 (17)	
CSS score (mean [SD])			
Total score	32.05 (±11.33)	37.22 (±12.03)	0.17[b]
Physical reactions	7.18 (±4.65)	8.78 (±3.65)	0.24[b]
Psychological reactions	11.23 (±4.87)	14.33 (±5.51)	0.07[b]
Psychophysiological reactions	7.18 (±3.84)	7.56 (±3.90)	0.76[b]
Psychological reactions with depressive components	6.45 (±3.92)	6.56 (±4.97)	0.94[b]
PCS-C score (mean [SD])			
Total score	25.05 (±9.89)	31.33 (±10.22)	0.06[b]
Rumination subscale	11.27 (±2.85)	12.61 (±2.72)	0.14[b]
Magnification subscale	3.95 (±3.03)	5.89 (±2.97)	0.05[b]
Helplessness subscale	9.82 (±5.27)	12.83 (±5.86)	0.09[b]

SD, standard deviation; CSS, Child Stress Scale (scores range from 0 to 140); PCS-C, Pain Catastrophizing Scale for Children (scores range from 0 to 52).
[a]χ^2 test.
[b]Student's *t*-test.

statistically significant interaction effects. Thus, the children who received the distraction intervention in the first puncture procedure benefited from the intervention in the second procedure with regard to pain assessment compared with children who did not receive the intervention during the first puncture procedure.

In addition, the analysis of carryover effects in the second painful procedure showed that 29% of the estimated effect in relieving pain, assessed by the

Table 4 Pain scores on FPS-R and VAS.

Pain assessment	Group 1 (AB) (n = 22)		Group 2 (BA) (n = 18)	
	A (Distraction)	B (No intervention)	A (Distraction)	B (No intervention)
FPS-R score (Mean ± SD)	1.91 ± 1.68	3.64 ± 3.06	2.67 ± 2.74	6.78 ± 3.15
VAS score (Mean ± SD)	1.50 ± 1.87	3.32 ± 3.42	2.33 ± 2.54	6.28 ± 2.86

SD, standard deviation.

FPS-R, was attributable to the distraction intervention.

3.5.2 Pain outcome in VAS measure

A statistically significant difference was found between the painful procedures with and without the distraction intervention. A significant effect of the intervention on pain relief was observed, with lower VAS scores for the painful procedures with distraction compared with no intervention ($F_{1,38}$ = 29.31, p < 0.001). The results also showed that the sequence of distraction exposure (AB vs. BA) in both groups ($F_{1,38}$ = 7.58, p = 0.01) and the first versus second painful procedure ($F_{1,38}$ = 3.99, p = 0.05) during which the distraction intervention was performed significantly affected the efficacy of audiovisual distraction. There were no statistically significant interaction effects. In addition, the analysis of carryover effects in the second period showed that 30% of the estimated effect in relieving pain, assessed by the VAS, was attributable to the distraction intervention.

4. Discussion

The results of this crossover study supported the initial hypothesis that audiovisual distraction reduced the perception of pain intensity during acute painful procedures in routine clinical practice in hospitalized children. The children in this study were patients who were hospitalized in a paediatric ward with different clinical diagnoses in different clinical specialties (e.g. onco-haematology, endocrinology, nephrology) and who were routinely prescribed venous or arterial puncture. Most of the children suffered from chronic disease and had low indicators of stress and pain catastrophizing prior to data collection.

The results showed that the sequence of exposure (AB or BA) and the first versus second puncture procedure during which the audiovisual distraction intervention was performed affected the perception of pain intensity. When the children received the distraction intervention during the first puncture procedure, pain scores were lower during the second painful procedure (without the distraction intervention) compared with pain scores in children who did not receive the distraction intervention during the first puncture procedure. These findings suggest that the children in G1 learned from the first distraction experience and utilized this cognitive resource during subsequent painful experience. These results indicate carryover effects during the seven consecutive days after the distraction procedure, in which approximately 30% of the estimated effects on pain relief were attributable to only the distraction intervention.

This study provides additional evidence on the efficacy of distraction interventions using a crossover design. The crossover design allowed us to expand investigations of the effects of distraction interventions by considering in the analysis the sequence of intervention exposure and the time at which it was applied. This methodology considers the individual as his own control, thus reducing confounding factors that may be associated with the studied phenomenon and consequently increasing the power of the reliability of the results (Senn, 2002).

Previous studies on audiovisual distraction also confirmed the efficacy of this intervention for pain relief in children. Wang et al. (2008) evaluated audiovisual distraction compared with a psychological intervention routine and control condition in children aged 8 and 9 years who had various paediatric diagnoses and underwent venipuncture for intravenous treatment. The findings showed that audiovisual distraction reduced the perception of pain intensity, improved patient compliance and resulted in an increase in the success rate of the procedures. Audiovisual distraction was also examined by Bellieni et al. (2006), who assessed the effects of this technique in school children who underwent blood draw compared with distraction that was performed by mothers using interactive toys. The study showed that audiovisual distraction was more effective than the distraction that was performed by the mothers, in which audiovisual distraction provided greater pain relief (indicated by the children's own accounts of pain intensity) and increased pain tolerance (indicated by the mothers' determination of pain scores in their children). The study differs from previous studies and provides additional information about the effect of audiovisual distraction on subsequent painful procedures. We also found a positive effect of audiovisual distraction for pain relief even in children with *a prior* history of pain.

Attentional processes may explain the effects of the distraction intervention in this study. The theory of limited attention capacity suggests that attention is a limited resource, in which attention that is given to one activity limits its concurrent availability for other activities. Following this proposal, the theory of multiple attentional resources suggests that the extent to which two competing activities use the same domain resources will dictate the extent to which they interfere or prevent the processing of others (Eccleston and Crombez, 1999; DeMore and Cohen, 2005; Johnson, 2005). The mechanism of distraction described by Eccleston and Crombez (1999) suggests that pain interrupts attentional demands, and the efficacy of distraction depends on the emotional meaning that is ascribed to a task such that the perception of pain can be changed.

Memory and learning processes may also aid the understanding of the effect of audiovisual distraction in subsequent painful procedures. Memory processes can explain the influence of prior painful experiences on future painful situations. Recent studies have shown that negative responses in episodes of pain are associated with non-adaptive future reactions (Kennedy et al., 2008; Noel et al., 2012). In addition, learning theory explains that a more immersive distracting activity lowers the pain that is experienced because of the greater child's self-efficacy to cope with the painful stimuli, and also the modelling process of distraction offer by health professionals (Chambers et al., 2002; Cohen, 2002; DeMore and Cohen, 2005; Page and Blanchette, 2009).

The main results of this study suggest that adopting distraction interventions during painful procedures may help children use more adaptive attitudes in future painful contexts. The clinical characteristics of this sample showed that most of the children had chronic diseases (indicated by the mean treatment time), likely resulting in trajectories of pain through the course of treatment, and were attended by different health services. Unfortunately, we had no specific information about the number of previous needle procedures that the children underwent in their trajectory. The findings strengthen the recommendation that distraction during acute painful procedures works even in children with a long-term history of pain, and who are exposed to prior painful procedures during the course of the disease.

Although the sample in this study was composed of children with various diagnoses and treatments, this study has ecological validity because the data were collected within the clinical context of a hospital. The present results may also be generalized to the extent to which the distraction technique provided pain relief, regardless of the children's diagnosis. Notably, the children in our study, although sick and hospitalized, had low levels of stress and pain catastrophizing. Such conditions may facilitate the effect of distraction for the relief of acute pain.

The present study has some limitations. The procedure was performed by the researcher and therefore did not have blinded conditions. The sample consisted of children with different diagnoses. The scale that evaluated pain catastrophizing was translated to Brazilian Portuguese, but the psychometric features of this translation have not been evaluated. The other outcomes that are usually associated with painful experiences, such as fear and distress, were not evaluated or controlled in this study. We also did not perform analyses to investigate whether there were any subgroups of children would be more or less helped by the intervention.

Regardless of these limitations, this study suggests that audiovisual distraction effectively relieved acute pain in hospitalized children. The findings indicate that audiovisual distraction may be an easy strategy to implement and inexpensive for effective pain relief in routine clinical procedures in hospitalized children. In this study, the psychologist performed the audiovisual distraction, although other health professionals may also be trained to apply this technique during painful procedures.

The current study also has implications for clinical practice. It provides evidence that the distraction technique for pain relief may be implemented in paediatric care contexts and improves the humane treatment of paediatric patients. This crossover clinical trial showed that the distraction intervention could be included in clinical protocols for pain management in paediatric units. Within the context of this study, we are currently updating the clinical protocol for pain management, and this distraction intervention will be included as a systematic routine during acute painful procedures.

Future studies should examine the effects of this distraction intervention during different painful procedures and its application by other health professionals. Other relevant outcomes, such as distress and fear, could also be investigated because they are associated with painful experiences. In addition, the efficacy of distraction should be examined based on the level of pain catastrophizing in children. Finally, other age groups of children, different clinical settings and carryover effects beyond 1 week should be examined to expand the power of generalization of these findings.

In conclusion, the present study contributes to the literature regarding the effects of distraction for acute pain management. The results indicate that audiovisual distraction is a powerful and effective non-pharmacological intervention for pain relief in paediatric inpatients. Moreover, the beneficial effects of this intervention were detected in subsequent acute painful procedures.

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Author contributions

N.C.A.C. Oliveira: conception and design of the study, data collection, analysis and interpretation of data and writing and revising the article. M.B.M. Linhares: conception and design of the study, analysis and interpretation of data and writing and revising the article. J.L.F. Santos: analysis and interpretation of data and revising the article. All authors approved this version of the article.

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